

AI ASSISTED CODING

Assignment-16.1

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Task 1 – Student Information System Schema

Task:

Design a database schema for a Student Information System and generate queries using AI.

Instructions:

- Tables: Students, Courses, Enrollments.
- Define primary keys, foreign keys, and relationships.
- Generate queries:

- o Insert a new student record.

- o Fetch all courses enrolled by a student.

- o Count number of students in each course.

Expected Output:

- SQL CREATE TABLE statements and queries for student–course relationships.

▼ Student Information System Schema

Below are the SQL `CREATE TABLE` statements for the `Students`, `Courses`, and `Enrollments` tables. These define the structure, primary keys, foreign keys, and relationships between the tables.

- **Students Table:** Stores basic student information.
- **Courses Table:** Stores details about the academic courses.
- **Enrollments Table:** Links students to courses, establishing a many-to-many relationship.

```
[11]
✓ Os  import sqlite3

conn = sqlite3.connect('student_system.db')
cursor = conn.cursor()

cursor.execute('''
CREATE TABLE Students (
    student_id INTEGER PRIMARY KEY AUTOINCREMENT,
    first_name TEXT NOT NULL,
    last_name TEXT NOT NULL,
    date_of_birth DATE,
    email TEXT UNIQUE NOT NULL
    .
```

```
);
''')

conn.commit()
conn.close()

print("Students table created.")
```

▼ Students table created.

```
[12]
✓ Os  import sqlite3

conn = sqlite3.connect('student_system.db')
cursor = conn.cursor()

cursor.execute('''
CREATE TABLE Courses (
    course_id INTEGER PRIMARY KEY AUTOINCREMENT,
    course_name TEXT NOT NULL,
    course_code TEXT UNIQUE NOT NULL,
    credits INTEGER NOT NULL
    );
''')
```

```
conn.commit()
conn.close()

print("Courses table created.")
```

Courses table created.

```
import sqlite3

conn = sqlite3.connect('student_system.db')
cursor = conn.cursor()

cursor.execute('''
CREATE TABLE Enrollments (
    enrollment_id INTEGER PRIMARY KEY AUTOINCREMENT,
    student_id INTEGER NOT NULL,
    course_id INTEGER NOT NULL,
    enrollment_date DATE NOT NULL,
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
    FOREIGN KEY (course_id) REFERENCES Courses(course_id)
);
''')

conn.commit()
conn.close()

print("Enrollments table created.")

... Enrollments table created.
```

Now that the tables are created, let's insert some sample data to make the queries demonstrate meaningful results.

```
import sqlite3

conn = sqlite3.connect('student_system.db')
cursor = conn.cursor()

cursor.execute("INSERT INTO Students (first_name, last_name, date_of_birth, email) VALUES ('AI")
cursor.execute("INSERT INTO Courses (course_name, course_code, credits) VALUES ('Introduction")
cursor.execute("INSERT INTO Enrollments (student_id, course_id, enrollment_date) VALUES (1, 1,")

conn.commit()
conn.close()

print("Sample data inserted.")

... Sample data inserted.
```

1. Insert a new student record

This query demonstrates how to add a new student to the `Students` table. Note that `student_id` is auto-incremented, so it's not explicitly provided.

```
15] import sqlite3

conn = sqlite3.connect('student_system.db')
cursor = conn.cursor()

cursor.execute("INSERT INTO Students (first_name, last_name, date_of_birth, email) VALUES ('Di
conn.commit()

cursor.execute("SELECT * FROM Students WHERE email = 'diana.prince@example.com';")
result = cursor.fetchone()
conn.close()

if result:
    print(result)
else:
    print("Student not found.")

... (4, 'Diana', 'Prince', '2000-08-10', 'diana.prince@example.com')
```

This query retrieves the `course_name` and `course_code` for all courses a student is enrolled in. You can change `student_id = 1` to query for a different student.

```
16] import sqlite3

conn = sqlite3.connect('student_system.db')
cursor = conn.cursor()

cursor.execute('''
SELECT
    S.first_name,
    S.last_name,
    C.course_name,
    C.course_code
FROM Students AS S
JOIN Enrollments AS E ON S.student_id = E.student_id
JOIN Courses AS C ON E.course_id = C.course_id
WHERE S.student_id = 1;
''')

results = cursor.fetchall()
conn.close()

for row in results:
    print(row)

... ('Alice', 'Smith', 'Introduction to Programming', 'CS101')
... ('Alice', 'Smith', 'Database Management', 'CS205')
```

of count the number of students in each course

This query lists each `course_name` along with the total number of students enrolled in it. Courses with no enrollments will not appear.

```
[17] ✓ 0s ▶ import sqlite3

conn = sqlite3.connect('student_system.db')
cursor = conn.cursor()

cursor.execute('''
SELECT
    C.course_name,
    COUNT(E.student_id) AS num_students_enrolled
FROM Courses AS C
LEFT JOIN Enrollments AS E ON C.course_id = E.course_id
GROUP BY C.course_name
ORDER BY num_students_enrolled DESC;
''')

results = cursor.fetchall()
conn.close()

for row in results:
    print(row)

... ('Introduction to Programming', 2)
    ('Database Management', 2)
    ('Calculus I', 1)
```

Task 2 – Hospital Management Database

Task:

Create schema and queries for a Hospital Management System.

Instructions:

- Tables: Doctors, Patients, Appointments.
- Use AI to define constraints (unique IDs, valid dates).
- Generate queries:
 - o List all appointments for a specific doctor.
 - o Retrieve patient history by patient ID.
 - o Count total patients treated by each doctor.

Expected Output:

- Normalized schema and SQL queries with joins.

▼ Hospital Management Database Schema

Below are the SQL `CREATE TABLE` statements for the `Doctors`, `Patients`, and `Appointments` tables, including primary keys, foreign keys, and other constraints. I'm using Python's `sqlite3` module to interact with the database.

- **Doctors Table:** Stores information about medical doctors.
- **Patients Table:** Stores details of patients.
- **Appointments Table:** Records appointments between doctors and patients, establishing relationships between the other two tables.

```
[18]
✓ Os
import sqlite3

conn = sqlite3.connect('hospital_management.db')
cursor = conn.cursor()

cursor.execute('''
CREATE TABLE Doctors (
    doctor_id INTEGER PRIMARY KEY AUTOINCREMENT,
    first_name TEXT NOT NULL,
    last_name TEXT NOT NULL,
    specialty TEXT NOT NULL,
    phone_number TEXT UNIQUE,
    email TEXT UNIQUE NOT NULL
);
''')

conn.commit()
conn.close()

print("Doctors table created.")
```

▼ ... Doctors table created.

```
9]
Os
import sqlite3

conn = sqlite3.connect('hospital_management.db')
cursor = conn.cursor()

cursor.execute('''
CREATE TABLE Patients (
    patient_id INTEGER PRIMARY KEY AUTOINCREMENT,
    first_name TEXT NOT NULL,
    last_name TEXT NOT NULL,
    date_of_birth DATE NOT NULL,
    gender TEXT,
    phone_number TEXT UNIQUE,
    email TEXT UNIQUE
);
''')

conn.commit()
conn.close()

print("Patients table created.")
```

... Patients table created.

[+ Code](#) [+ Text](#)

```
[20]
✓ Os
import sqlite3

conn = sqlite3.connect('hospital_management.db')
cursor = conn.cursor()

cursor.execute('''
CREATE TABLE Appointments (
    appointment_id INTEGER PRIMARY KEY AUTOINCREMENT,
    doctor_id INTEGER NOT NULL,
    patient_id INTEGER NOT NULL,
    appointment_date DATE NOT NULL,
    appointment_time TIME NOT NULL,
    reason TEXT,
    FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id),
    FOREIGN KEY (patient_id) REFERENCES Patients(patient_id)
);
''')

conn.commit()
conn.close()

print("Appointments table created.")
```

▼ Appointments table created.

▼ Insert Sample Data

Now, let's populate our tables with some sample data to make the queries more illustrative.

```
[21] 1 import sqlite3
    2
    3 conn = sqlite3.connect('hospital_management.db')
    4 cursor = conn.cursor()
    5
    6 # Insert Doctors
    7 cursor.execute("INSERT INTO Doctors (first_name, last_name, specialty, phone_number, email) VALUES ('John', 'Doe', 'Cardiology', '555-1234')")
    8 cursor.execute("INSERT INTO Doctors (first_name, last_name, specialty, phone_number, email) VALUES ('Jane', 'Smith', 'Pediatrics', '555-5678')")
    9 cursor.execute("INSERT INTO Doctors (first_name, last_name, specialty, phone_number, email) VALUES ('Alice', 'Brown', 'Dermatology', '555-9012')")
   10
   11 # Insert Patients
   12 cursor.execute("INSERT INTO Patients (first_name, last_name, date_of_birth, gender, phone_number, email) VALUES ('Robert', 'White', '1980-01-15', 'M', '555-2345')")
   13 cursor.execute("INSERT INTO Patients (first_name, last_name, date_of_birth, gender, phone_number, email) VALUES ('Emily', 'Green', '1990-03-22', 'F', '555-6789')")
   14 cursor.execute("INSERT INTO Patients (first_name, last_name, date_of_birth, gender, phone_number, email) VALUES ('Michael', 'Black', '1975-07-08', 'M', '555-3456')")
   15
   16 # Insert Appointments
   17 cursor.execute("INSERT INTO Appointments (doctor_id, patient_id, appointment_date, appointment_time, reason) VALUES (1, 1, '2023-10-26', '10:00', 'Chest Pain')")
   18 cursor.execute("INSERT INTO Appointments (doctor_id, patient_id, appointment_date, appointment_time, reason) VALUES (1, 2, '2023-10-26', '11:00', 'Follow-up')")
   19 cursor.execute("INSERT INTO Appointments (doctor_id, patient_id, appointment_date, appointment_time, reason) VALUES (2, 2, '2023-10-27', '09:00', 'Allergy Test')")
   20 cursor.execute("INSERT INTO Appointments (doctor_id, patient_id, appointment_date, appointment_time, reason) VALUES (3, 1, '2023-10-28', '14:00', 'Skin Check')")
   21 cursor.execute("INSERT INTO Appointments (doctor_id, patient_id, appointment_date, appointment_time, reason) VALUES (1, 3, '2023-10-29', '15:00', 'Blood Pressure Check')")
   22
   23 conn.commit()
   24 conn.close()
   25
   26 print("Sample data inserted.")
```

```
[22] 1 import sqlite3
    2
    3 conn = sqlite3.connect('hospital_management.db')
    4 cursor = conn.cursor()
    5
    6 cursor.execute('''
    7     SELECT
    8         D.first_name AS DoctorFirstName,
    9         D.last_name AS DoctorLastName,
   10         P.first_name AS PatientFirstName,
   11         P.last_name AS PatientLastName,
   12         A.appointment_date,
   13         A.appointment_time,
   14         A.reason
   15     FROM Appointments AS A
   16     JOIN Doctors AS D ON A.doctor_id = D.doctor_id
   17     JOIN Patients AS P ON A.patient_id = P.patient_id
   18     WHERE D.doctor_id = 1;
   19 ''')
   20
   21 results = cursor.fetchall()
   22 conn.close()
   23
   24 print(f"Appointments for Doctor John Doe (ID=1):")
   25 if results:
   26     for row in results:
   27         print(row)
   28 else:
   29     print("No appointments found.")
   30
   31 ... Appointments for Doctor John Doe (ID=1):
   32 ('John', 'Doe', 'Robert', 'White', '2023-10-26', '10:00', 'Chest Pain')
   33 ('John', 'Doe', 'Emily', 'Green', '2023-10-26', '11:00', 'Follow-up')
   34 ('John', 'Doe', 'Michael', 'Black', '2023-10-29', '15:00', 'Blood Pressure Check')
```

demonstrates a patient's historical interactions within the system.

```
import sqlite3

conn = sqlite3.connect('hospital_management.db')
cursor = conn.cursor()

cursor.execute('''
SELECT
    P.first_name AS PatientFirstName,
    P.last_name AS PatientLastName,
    A.appointment_date,
    A.appointment_time,
    D.first_name AS DoctorFirstName,
    D.last_name AS DoctorLastName,
    A.reason
FROM Appointments AS A
JOIN Patients AS P ON A.patient_id = P.patient_id
JOIN Doctors AS D ON A.doctor_id = D.doctor_id
WHERE P.patient_id = 2
ORDER BY A.appointment_date DESC, A.appointment_time DESC;
''')

results = cursor.fetchall()
conn.close()

print(f"Appointment history for Patient Emily Green (ID=2):")
if results:
    for row in results:
        print(row)
else:
    print("No history found.")
```

... Appointment history for Patient Emily Green (ID=2):
(('Emily', 'Green', '2023-10-27', '09:30', 'Jane', 'Smith', 'Child Checkup'))
(('Emily', 'Green', '2023-10-26', '11:00', 'John', 'Doe', 'Follow-up'))

3. Count total patients treated by each doctor

This query aggregates the number of unique patients associated with each doctor, providing insight into each doctor's patient load. It joins the `Doctors` and `Appointments` tables and counts distinct `patient_id`s.

```
import sqlite3

conn = sqlite3.connect('hospital_management.db')
cursor = conn.cursor()

cursor.execute('''
SELECT
    D.first_name,
    D.last_name,
    D.specialty,
    COUNT(DISTINCT A.patient_id) AS total_unique_patients
FROM Doctors AS D
LEFT JOIN Appointments AS A ON D.doctor_id = A.doctor_id
GROUP BY D.doctor_id
ORDER BY total_unique_patients DESC, D.last_name, D.first_name;
''')

results = cursor.fetchall()
conn.close()

print("Total unique patients treated by each doctor:")
if results:
    for row in results:
        print(row)
else:
    print("No doctors found or no patients treated.")
```

... Total unique patients treated by each doctor:
(('John', 'Doe', 'Cardiology', 3))
(('Alice', 'Brown', 'Dermatology', 1))
(('Jane', 'Smith', 'Pediatrics', 1))

Task 3 – Library Database

Task:

Design schema for a Library Management System.

Instructions:

- Tables: Books, Members, Loans.
- Use AI to suggest indexing strategy for faster queries.
- Generate queries:
 - o Retrieve all books currently issued.
 - o Find overdue books (loan date > 30 days).
 - o Count number of books loaned by each member.

Expected Output:

- Schema + SQL queries demonstrating joins and conditions.

```
Library Management System Schema

Below are the SQL CREATE TABLE statements for the Books, Members, and Loans tables, including primary keys, foreign keys, and other constraints. I'm using Python's sqlite3 module to interact with the database.

• Books Table: Stores details about the books in the library.
• Members Table: Stores information about library members.
• Loans Table: Records each instance of a book being loaned to a member.

[25] import sqlite3
✓ Os conn = sqlite3.connect('library_management.db')
cursor = conn.cursor()

cursor.execute('''
CREATE TABLE Books (
    book_id INTEGER PRIMARY KEY AUTOINCREMENT,
    title TEXT NOT NULL,
    author TEXT NOT NULL,
    isbn TEXT UNIQUE NOT NULL,
    publication_year INTEGER,
    is_available BOOLEAN NOT NULL DEFAULT 1 -- 1 for available, 0 for loaned
);
''')

conn.commit()
conn.close()

print("Books table created.")

Books table created.

import sqlite3

conn = sqlite3.connect('library_management.db')
cursor = conn.cursor()

cursor.execute('''
CREATE TABLE Members (
    member_id INTEGER PRIMARY KEY AUTOINCREMENT,
    first_name TEXT NOT NULL,
    last_name TEXT NOT NULL,
    email TEXT UNIQUE NOT NULL,
    registration_date DATE NOT NULL
);
''')

conn.commit()
conn.close()

print("Members table created.")

Members table created.
```

```
[27]
✓ 0s
import sqlite3

conn = sqlite3.connect('library_management.db')
cursor = conn.cursor()

cursor.execute('''
CREATE TABLE Loans (
    loan_id INTEGER PRIMARY KEY AUTOINCREMENT,
    book_id INTEGER NOT NULL,
    member_id INTEGER NOT NULL,
    loan_date DATE NOT NULL,
    due_date DATE NOT NULL,
    return_date DATE, -- NULL if not yet returned Loading...
    FOREIGN KEY (book_id) REFERENCES Books(book_id),
    FOREIGN KEY (member_id) REFERENCES Members(member_id)
);
''')

conn.commit()
conn.close()

print("Loans table created.")

... Loans table created.
```

Suggested Indexing Strategy

To improve the performance of queries, especially those involving `WHERE` clauses, `JOIN` operations, and `ORDER BY` clauses, it's beneficial to create indexes on frequently accessed columns. For this Library Management System:

- **Books.isbn** : Often used for quick lookups of a specific book. It's already `UNIQUE` which often implies an index, but explicitly adding one can ensure optimal performance for searches.
- **Loans.book_id** : This is a foreign key and will be frequently used in `JOIN` operations when retrieving loan information related to a specific book.
- **Loans.member_id** : This is a foreign key and will be frequently used in `JOIN` operations when retrieving loan information related to a specific member.
- **Loans.return_date** : This column will be frequently queried to find currently issued books (where `return_date IS NULL`) or to check return history.
- **Loans.loan_date** : This column will be used for queries involving loan periods and overdue books.

Here are the SQL statements to create these indexes:

```
import sqlite3

conn = sqlite3.connect('library_management.db')
cursor = conn.cursor()

cursor.execute('CREATE INDEX idx_books_isbn ON Books (isbn);')
cursor.execute('CREATE INDEX idx_loans_book_id ON Loans (book_id);')
cursor.execute('CREATE INDEX idx_loans_member_id ON Loans (member_id);')
cursor.execute('CREATE INDEX idx_loans_return_date ON Loans (return_date);')
cursor.execute('CREATE INDEX idx_loans_loan_date ON Loans (loan_date);')

conn.commit()
conn.close()

print("Indexes created.")

Indexes created.
```

```

import sqlite3
from datetime import date, timedelta

conn = sqlite3.connect('library_management.db')
cursor = conn.cursor()

# Insert Books
cursor.execute("INSERT INTO Books (title, author, isbn, publication_year, is_available) VALUES (?, ?, ?, ?, ?);", ('The Hitchhiker\'s Guide to the Galaxy', 'Douglas Adams', '0312864641', 1979, 1))
cursor.execute("INSERT INTO Books (title, author, isbn, publication_year, is_available) VALUES (?, ?, ?, ?, ?);", ('Pride and Prejudice', 'Jane Austen', '0141439528', 1813, 1))
cursor.execute("INSERT INTO Books (title, author, isbn, publication_year, is_available) VALUES (?, ?, ?, ?, ?);", ('1984', 'George Orwell', '0451526342', 1949, 1))
cursor.execute("INSERT INTO Books (title, author, isbn, publication_year, is_available) VALUES (?, ?, ?, ?, ?);", ('To Kill a Mockingbird', 'Harper Lee', '0061120081', 1960, 1))

# Insert Members
cursor.execute("INSERT INTO Members (first_name, last_name, email, registration_date) VALUES (?, ?, ?, ?);", ('Arthur', 'Dent', 'arthur.dent@hitchhiker.com', date(1979, 1, 1)))
cursor.execute("INSERT INTO Members (first_name, last_name, email, registration_date) VALUES (?, ?, ?, ?);", ('Elizabeth', 'Bennet', 'elizabeth.bennet@prideandprejudice.com', date(1813, 1, 1)))
cursor.execute("INSERT INTO Members (first_name, last_name, email, registration_date) VALUES (?, ?, ?, ?);", ('Winston', 'Smith', 'winston.smith@1984.com', date(1949, 1, 1)))

# Insert Loans
today = date.today()
loan_date_1 = today - timedelta(days=5)
loan_date_2 = today - timedelta(days=40) # Overdue example
loan_date_3 = today - timedelta(days=10)
loan_date_4 = today - timedelta(days=2)

due_date_1 = loan_date_1 + timedelta(days=14)
due_date_2 = loan_date_2 + timedelta(days=14)
due_date_3 = loan_date_3 + timedelta(days=14)
due_date_4 = loan_date_4 + timedelta(days=14)

# Book 1 loaned to Member 1 (currently issued)
cursor.execute("INSERT INTO Loans (book_id, member_id, loan_date, due_date, return_date) VALUES (?, ?, ?, ?, ?);", (1, 1, loan_date_1, due_date_1, None))
# Book 3 loaned to Member 3 (overdue)
cursor.execute("INSERT INTO Loans (book_id, member_id, loan_date, due_date, return_date) VALUES (?, ?, ?, ?, ?);", (3, 3, loan_date_2, due_date_2, None))
# Book 1 loaned to Member 2 (returned)
cursor.execute("INSERT INTO Loans (book_id, member_id, loan_date, due_date, return_date) VALUES (?, ?, ?, ?, ?);", (1, 2, loan_date_3, due_date_3, loan_date_3))
# Book 4 loaned to Member 1 (currently issued, not overdue)

```

```

import sqlite3

conn = sqlite3.connect('library_management.db')
cursor = conn.cursor()

cursor.execute('''
SELECT
    B.title,
    B.author,
    M.first_name AS LoanedToFirstName,
    M.last_name AS LoanedToLastName,
    L.loan_date,
    L.due_date
FROM Loans AS L
JOIN Books AS B ON L.book_id = B.book_id
JOIN Members AS M ON L.member_id = M.member_id
WHERE L.return_date IS NULL;
''')

results = cursor.fetchall()
conn.close()

print("Books currently issued:")
if results:
    for row in results:
        print(row)
else:
    print("No books currently issued.")

```

```
import sqlite3

conn = sqlite3.connect('library_management.db')
cursor = conn.cursor()

cursor.execute('''
SELECT
    B.title,
    B.author,
    M.first_name AS LoanedToFirstName,
    M.last_name AS LoanedToLastName,
    L.loan_date,
    L.due_date
FROM Loans AS L
JOIN Books AS B ON L.book_id = B.book_id
JOIN Members AS M ON L.member_id = M.member_id
WHERE L.return_date IS NULL AND L.due_date < DATE('now');
''')

results = cursor.fetchall()
conn.close()

print("Overdue books:")
if results:
    for row in results:
        print(row)
else:
    print("No overdue books found.")

Overdue books:
('1984', 'George Orwell', 'Winston', 'Smith', '2026-02-04', '2026-02-18')
```

```
import sqlite3

conn = sqlite3.connect('library_management.db')
cursor = conn.cursor()

cursor.execute('''
SELECT
    M.first_name,
    M.last_name,
    COUNT(L.loan_id) AS total_books_loaned
FROM Members AS M
LEFT JOIN Loans AS L ON M.member_id = L.member_id
GROUP BY M.member_id, M.first_name, M.last_name
ORDER BY total_books_loaned DESC, M.last_name, M.first_name;
''')

results = cursor.fetchall()
conn.close()

print("Number of books loaned by each member:")
if results:
    for row in results:
        print(row)
else:
    print("No members found or no books loaned.")

... Number of books loaned by each member:
('Arthur', 'Dent', 2)
('Elizabeth', 'Bennet', 1)
('Winston', 'Smith', 1)
```

Task 4 – Real-Time Application: Online Shopping Database

Scenario:

Design a database for an E-commerce platform.

Requirements:

- Tables: Users, Products, Orders, OrderDetails.
- Generate queries to:
 - o Retrieve all orders by a user.
 - o Find the most purchased product.

o Calculate total revenue in a given month.

- AI should also suggest normalization improvements and query optimization.

Expected Output:

- Complete schema with relationships + SQL queries for analytics.

```

  E-commerce Platform Database Schema

  Below are the SQL CREATE TABLE statements for the Users, Products, Orders, and OrderDetails tables. These define the
  structure, primary keys, foreign keys, and relationships between the tables.

  • Users Table: Stores customer information.
  • Products Table: Stores details of items available for sale.
  • Orders Table: Records each customer order.
  • OrderDetails Table: Links products to specific orders and specifies quantities and prices at the time of purchase.

[38]
✓ Os ▶ import sqlite3

      conn = sqlite3.connect('e_commerce.db')
      cursor = conn.cursor()

      cursor.execute('''
      CREATE TABLE Users (
        user_id INTEGER PRIMARY KEY AUTOINCREMENT,
        first_name TEXT NOT NULL,
        last_name TEXT NOT NULL,
        email TEXT UNIQUE NOT NULL,
        registration_date DATE NOT NULL
      );
      ''')

      conn.commit()
      conn.close()

      print("Users table created.")

  ... Users table created.

9]
Os ▶ import sqlite3

      conn = sqlite3.connect('e_commerce.db')
      cursor = conn.cursor()

      cursor.execute('''
      CREATE TABLE Products (
        product_id INTEGER PRIMARY KEY AUTOINCREMENT,
        product_name TEXT UNIQUE NOT NULL,
        description TEXT,
        price REAL NOT NULL,
        stock_quantity INTEGER NOT NULL
      );
      ''')

      conn.commit()
      conn.close()

      print("Products table created.")

  ... Products table created.
```

```
import sqlite3

conn = sqlite3.connect('e_commerce.db')
cursor = conn.cursor()

cursor.execute('''
CREATE TABLE Orders (
    order_id INTEGER PRIMARY KEY AUTOINCREMENT,
    user_id INTEGER NOT NULL,
    order_date DATE NOT NULL,
    total_amount REAL NOT NULL,
    status TEXT NOT NULL DEFAULT 'pending',
    FOREIGN KEY (user_id) REFERENCES Users(user_id)
);
''')

conn.commit()
conn.close()

print("Orders table created.")
```

```
import sqlite3

conn = sqlite3.connect('e_commerce.db')
cursor = conn.cursor()

cursor.execute('''
CREATE TABLE OrderDetails (
    order_detail_id INTEGER PRIMARY KEY AUTOINCREMENT,
    order_id INTEGER NOT NULL,
    product_id INTEGER NOT NULL,
    quantity INTEGER NOT NULL,
    price_at_purchase REAL NOT NULL, -- Price might change over time, so record it
    FOREIGN KEY (order_id) REFERENCES Orders(order_id),
    FOREIGN KEY (product_id) REFERENCES Products(product_id)
);
''')

conn.commit()
conn.close()

print("OrderDetails table created.")
```

... OrderDetails table created.

Normalization Improvements

The current schema is designed to be in at least 3rd Normal Form (3NF). Here's a breakdown of its normalization and potential improvements:

- **1NF (First Normal Form):** All tables have primary keys, and there are no repeating groups or multi-valued attributes. Each column contains atomic values.
- **2NF (Second Normal Form):** All non-key attributes are fully functionally dependent on the primary key. For example, in `OrderDetails`, `quantity` and `price_at_purchase` depend on `order_detail_id` (the primary key).
- **3NF (Third Normal Form):** There are no transitive dependencies. Non-key attributes do not depend on other non-key attributes.
 - For instance, `total_amount` in `Orders` is a calculated field (sum of `quantity * price_at_purchase` from `OrderDetails`). Storing it here technically violates 3NF (as it depends on `OrderDetails` data, not directly on `order_id` in a non-transitive way). However, it's a common denormalization for performance (to avoid recalculating total for every order view) and consistency (capturing the total at the time of order). If strict 3NF is required, `total_amount` could be removed and calculated on the fly.

Potential Further Normalization (or Considerations):

- **Address Table:** For a real e-commerce platform, `Users` might need separate shipping and billing addresses. These could be moved to a dedicated `Addresses` table, linked to `Users` by `user_id`.
- **Categories Table:** `Products` could have a `category_id` linking to a `Categories` table (`category_id`, `category_name`, `description`) for better organization and searchability.
- **Reviews/Ratings Table:** A separate table for customer reviews and ratings of products (`review_id`, `product_id`, `user_id`, `rating`, `comment`, `review_date`).

Query Optimization Suggestions

To ensure faster query performance, especially with growing data, adding indexes is crucial:

- **Foreign Keys:** Always index foreign key columns (`user_id` in `Orders`, `order_id` and `product_id` in `OrderDetails`) as they are frequently used in `JOIN` operations.
 - `CREATE INDEX idx_orders_user_id ON Orders (user_id);`
 - `CREATE INDEX idx_order_details_order_id ON OrderDetails (order_id);`
 - `CREATE INDEX idx_order_details_product_id ON OrderDetails (product_id);`
- **Date Columns:** For queries involving date ranges (e.g., total revenue in a month), index `order_date` in the `Orders` table.
 - `CREATE INDEX idx_orders_order_date ON Orders (order_date);`
- **Frequently Filtered/Sorted Columns:** If `product_name` or `product_id` are often used in `WHERE` or `ORDER BY` clauses independently, ensure they are indexed. `product_id` is already a primary key, which typically comes with an implicit index. `product_name` is `UNIQUE` which also implies an index.

Let's apply these indexes:

```
import sqlite3

conn = sqlite3.connect('e_commerce.db')
cursor = conn.cursor()

cursor.execute('CREATE INDEX idx_orders_user_id ON Orders (user_id);')
cursor.execute('CREATE INDEX idx_order_details_order_id ON OrderDetails (order_id);')
cursor.execute('CREATE INDEX idx_order_details_product_id ON OrderDetails (product_id);')
cursor.execute('CREATE INDEX idx_orders_order_date ON Orders (order_date);')

conn.commit()
conn.close()

print("Indexes created.")
```

Indexes created.

Insert Sample Data

Let's populate our tables with some sample data to demonstrate the queries and relationships.

```
import sqlite3
from datetime import date

conn = sqlite3.connect('e_commerce.db')
cursor = conn.cursor()

# Insert Users
cursor.execute("INSERT INTO Users (first_name, last_name, email, registration_date) VALUES (?, ?, ?, ?);", ('John', 'Doe', 'john.doe@example.com', '2023-01-10'))
cursor.execute("INSERT INTO Users (first_name, last_name, email, registration_date) VALUES (?, ?, ?, ?);", ('Jane', 'Smith', 'jane.smith@example.com', '2023-02-15'))
cursor.execute("INSERT INTO Users (first_name, last_name, email, registration_date) VALUES (?, ?, ?, ?);", ('Alice', 'Brown', 'alice.brown@example.com', '2023-03-01'))

# Insert Products
cursor.execute("INSERT INTO Products (product_name, description, price, stock_quantity) VALUES (?, ?, ?, ?);", ('Laptop Pro', 'High-performance laptop', 1200.00, 50))
cursor.execute("INSERT INTO Products (product_name, description, price, stock_quantity) VALUES (?, ?, ?, ?);", ('Wireless Mouse', 'Ergonomic wireless mouse', 25.00, 200))
cursor.execute("INSERT INTO Products (product_name, description, price, stock_quantity) VALUES (?, ?, ?, ?);", ('Mechanical Keyboard', 'RGB Mechanical Keyboard', 80.00, 100))
cursor.execute("INSERT INTO Products (product_name, description, price, stock_quantity) VALUES (?, ?, ?, ?);", ('Monitor Ultrawide', '34-inch ultrawide monitor', 450.00, 30))

# Insert Orders
# Order 1 by John Doe (user_id 1)
cursor.execute("INSERT INTO Orders (user_id, order_date, total_amount, status) VALUES (?, ?, ?, ?);", (1, '2024-03-05', 1225.00, 'completed')) # 1 Laptop + 1 Mouse
# Order 2 by Jane Smith (user_id 2)
cursor.execute("INSERT INTO Orders (user_id, order_date, total_amount, status) VALUES (?, ?, ?, ?);", (2, '2024-03-10', 555.00, 'processing')) # 1 Monitor + 1 Keyboard + 1 Mouse
# Order 3 by John Doe (user_id 1)
cursor.execute("INSERT INTO Orders (user_id, order_date, total_amount, status) VALUES (?, ?, ?, ?);", (1, '2024-04-01', 80.00, 'pending')) # 1 Keyboard
# Order 4 by Jane Smith (user_id 2)
cursor.execute("INSERT INTO Orders (user_id, order_date, total_amount, status) VALUES (?, ?, ?, ?);", (2, '2024-04-02', 1200.00, 'completed')) # 1 Laptop

# Insert OrderDetails
# Order 1 details
cursor.execute("INSERT INTO OrderDetails (order_id, product_id, quantity, price_at_purchase) VALUES (?, ?, ?, ?);", (1, 1, 1, 1200.00))
cursor.execute("INSERT INTO OrderDetails (order_id, product_id, quantity, price_at_purchase) VALUES (?, ?, ?, ?);", (1, 2, 1, 25.00))
# Order 2 details
cursor.execute("INSERT INTO OrderDetails (order_id, product_id, quantity, price_at_purchase) VALUES (?, ?, ?, ?);", (2, 4, 1, 450.00))
cursor.execute("INSERT INTO OrderDetails (order_id, product_id, quantity, price_at_purchase) VALUES (?, ?, ?, ?);", (2, 3, 1, 80.00))
cursor.execute("INSERT INTO OrderDetails (order_id, product_id, quantity, price_at_purchase) VALUES (?, ?, ?, ?);", (2, 2, 1, 25.00))
# Order 3 details
cursor.execute("INSERT INTO OrderDetails (order_id, product_id, quantity, price_at_purchase) VALUES (?, ?, ?, ?);", (3, 3, 1, 80.00))
# Order 4 details
cursor.execute("INSERT INTO OrderDetails (order_id, product_id, quantity, price_at_purchase) VALUES (?, ?, ?, ?);", (4, 1, 1, 1200.00))

conn.commit()
conn.close()

print("Sample data inserted.")
```

Sample data inserted.

```
[44]
✓ Os
conn = sqlite3.connect('e_commerce.db')
cursor = conn.cursor()

cursor.execute('''
SELECT
    U.first_name || ' ' || U.last_name AS UserName,
    O.order_id,
    O.order_date,
    O.total_amount,
    O.status,
    P.product_name,
    OD.quantity,
    OD.price_at_purchase
FROM Users AS U
JOIN Orders AS O ON U.user_id = O.user_id
JOIN OrderDetails AS OD ON O.order_id = OD.order_id
JOIN Products AS P ON OD.product_id = P.product_id
WHERE U.user_id = 1
ORDER BY O.order_date DESC, O.order_id, P.product_name;
''')

results = cursor.fetchall()
conn.close()

print("All orders for User ID 1 (John Doe):")
if results:
    for row in results:
        print(row)
else:
    print("No orders found for this user.")

... All orders for User ID 1 (John Doe):
('John Doe', 3, '2024-04-01', 80.0, 'pending', 'Mechanical Keyboard', 1, 80.0)
('John Doe', 1, '2024-03-05', 1225.0, 'completed', 'Laptop Pro', 1, 1200.0)
('John Doe', 1, '2024-03-05', 1225.0, 'completed', 'Wireless Mouse', 1, 25.0)
```

2. Find the most purchased product

This query determines which product has been purchased the most across all orders by summing the `quantity` in `OrderDetails`, grouped by `product_id` and `product_name`.

```
[45]
✓ Os
import sqlite3

conn = sqlite3.connect('e_commerce.db')
cursor = conn.cursor()

cursor.execute('''
SELECT
    P.product_name,
    SUM(OD.quantity) AS total_quantity_purchased
FROM OrderDetails AS OD
JOIN Products AS P ON OD.product_id = P.product_id
GROUP BY P.product_name
ORDER BY total_quantity_purchased DESC
LIMIT 1;
''')

result = cursor.fetchone()
conn.close()

print("Most purchased product:")
if result:
    print(result)
else:
    print("No products purchased yet.")

... Most purchased product:
('Laptop Pro', 2)
```


3. Calculate total revenue in a given month (e.g., March 2024)

This query calculates the total revenue from completed orders within a specified month. It sums the `total_amount` from the `Orders` table, filtering by `order_date` and `status`.

```
[46] import sqlite3
✓ Os conn = sqlite3.connect('e_commerce.db')
      cursor = conn.cursor()

      # Define the month for which to calculate revenue
      # For example, March 2024
      start_date = '2024-03-01'
      end_date = '2024-03-31'

      cursor.execute('''
      SELECT
          SUM(total_amount) AS total_revenue
      FROM Orders
      WHERE order_date BETWEEN ? AND ?
            AND status = 'completed';
      ''', (start_date, end_date))

      result = cursor.fetchone()
      conn.close()

      print(f"Total revenue for orders completed between {start_date} and {end_date}:")
      if result and result[0] is not None:
          print(f"${result[0]:.2f}")
      else:
          print("No revenue found for this period.")
```

*** Total revenue for orders completed between 2024-03-01 and 2024-03-31:
\$1,225.00

Task 5 – University Examination Database

Design a database schema for a University Examination System and generate SQL queries using AI.

Instructions:

- Tables: Students, Subjects, Exams, Results.
- Define primary keys, foreign keys, and referential integrity constraints.
- Generate queries:
 - o Insert examination results for a student.
 - o Retrieve subject-wise marks for a student.
 - o Calculate average marks for each subject.

Expected Output:

- Normalized schema and SQL queries involving aggregation and joins.

University Examination System Database Schema

Below are the SQL `CREATE TABLE` statements for the `Students`, `Subjects`, `Exams`, and `Results` tables. These define the structure, primary keys, foreign keys, and referential integrity between the tables. I'm using Python's `sqlite3` module to interact with the database.

```
[47]  
✓ Os  
import sqlite3  
  
conn = sqlite3.connect('university_examination.db')  
cursor = conn.cursor()  
  
cursor.execute('''  
CREATE TABLE Students (  
    student_id INTEGER PRIMARY KEY AUTOINCREMENT,  
    first_name TEXT NOT NULL,  
    last_name TEXT NOT NULL,  
    date_of_birth DATE,  
    enrollment_date DATE NOT NULL,  
    email TEXT UNIQUE NOT NULL  
);  
''')  
  
conn.commit()  
conn.close()  
  
print("Students table created.")
```

... Students table created.

```
import sqlite3  
  
conn = sqlite3.connect('university_examination.db')  
cursor = conn.cursor()  
  
cursor.execute('''  
CREATE TABLE Subjects (  
    subject_id INTEGER PRIMARY KEY AUTOINCREMENT,  
    subject_name TEXT UNIQUE NOT NULL,  
    subject_code TEXT UNIQUE NOT NULL,  
    credits INTEGER NOT NULL  
);  
''')  
  
conn.commit()  
conn.close()  
  
print("Subjects table created.")
```

... Subjects table created.

```
import sqlite3  
  
conn = sqlite3.connect('university_examination.db')  
cursor = conn.cursor()  
  
cursor.execute('''  
CREATE TABLE Exams (  
    exam_id INTEGER PRIMARY KEY AUTOINCREMENT,  
    subject_id INTEGER NOT NULL,  
    exam_date DATE NOT NULL,  
    exam_type TEXT NOT NULL, -- e.g., 'Midterm', 'Final', 'Quiz'  
    max_marks INTEGER NOT NULL,  
    FOREIGN KEY (subject_id) REFERENCES Subjects(subject_id)  
);  
''')  
  
conn.commit()  
conn.close()  
  
print("Exams table created.")
```

... Exams table created.

```
import sqlite3

conn = sqlite3.connect('university_examination.db')
cursor = conn.cursor()

cursor.execute('''
CREATE TABLE Results (
    result_id INTEGER PRIMARY KEY AUTOINCREMENT,
    student_id INTEGER NOT NULL,
    exam_id INTEGER NOT NULL,
    marks_obtained INTEGER NOT NULL,
    grade TEXT, -- Optional: A, B, C, etc.
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
    FOREIGN KEY (exam_id) REFERENCES Exams(exam_id)
);
''')

conn.commit()
conn.close()

print("Results table created.")
```

Results table created.

▼ Insert Sample Data

Let's populate our tables with some sample data to demonstrate the queries.

```
[51]
✓ Os  import sqlite3

conn = sqlite3.connect('university_examination.db')
cursor = conn.cursor()

# Insert Students
cursor.execute("INSERT INTO Students (first_name, last_name, date_of_birth, enrollment_date, email) VALUES (?, ?, ?, ?, ?);", ('Anna', 'Lee', '2003-01-15', '2023-09-01', 'anna.lee@university.edu'))
cursor.execute("INSERT INTO Students (first_name, last_name, date_of_birth, enrollment_date, email) VALUES (?, ?, ?, ?, ?);", ('Ben', 'Carter', '2003-03-22', '2023-09-01', 'ben.carter@university.edu'))
cursor.execute("INSERT INTO Students (first_name, last_name, date_of_birth, enrollment_date, email) VALUES (?, ?, ?, ?, ?);", ('Chloe', 'Davis', '2003-05-10', '2023-09-01', 'chloe.davis@university.edu'))

# Insert Subjects
cursor.execute("INSERT INTO Subjects (subject_name, subject_code, credits) VALUES (?, ?, ?);", ('Calculus I', 'MATH101', 3))
cursor.execute("INSERT INTO Subjects (subject_name, subject_code, credits) VALUES (?, ?, ?);", ('Introduction to Programming', 'CS101', 3))
cursor.execute("INSERT INTO Subjects (subject_name, subject_code, credits) VALUES (?, ?, ?);", ('Physics I', 'PHY101', 3))

# Insert Exams
cursor.execute("INSERT INTO Exams (subject_id, exam_date, exam_type, max_marks) VALUES (?, ?, ?, ?);", (1, '2023-10-15', 'Midterm', 100))
cursor.execute("INSERT INTO Exams (subject_id, exam_date, exam_type, max_marks) VALUES (?, ?, ?, ?);", (2, '2023-10-20', 'Midterm', 100))
cursor.execute("INSERT INTO Exams (subject_id, exam_date, exam_type, max_marks) VALUES (?, ?, ?, ?);", (1, '2023-12-10', 'Final', 100))
cursor.execute("INSERT INTO Exams (subject_id, exam_date, exam_type, max_marks) VALUES (?, ?, ?, ?);", (2, '2023-12-15', 'Final', 100))
cursor.execute("INSERT INTO Exams (subject_id, exam_date, exam_type, max_marks) VALUES (?, ?, ?, ?);", (3, '2023-11-05', 'Midterm', 100))

conn.commit()
conn.close()

print("Sample data inserted (Students, Subjects, Exams).")
```

... Sample data inserted (Students, Subjects, Exams).

▼ 1. Insert examination results for a student

This query demonstrates how to add a new result record to the `Results` table for a specific student and exam. We'll insert results for `Student Anna Lee` (ID=1) for `Exams 1` (Calculus Midterm) and `2` (Programming Midterm) and for `Student Ben Carter` (ID=2) for `Exam 1`.

```
[52]
✓ Os  import sqlite3

conn = sqlite3.connect('university_examination.db')
cursor = conn.cursor()

# Insert results for Anna Lee (student_id 1)
cursor.execute("INSERT INTO Results (student_id, exam_id, marks_obtained, grade) VALUES (?, ?, ?, ?);", (1, 1, 85, 'A')) # Anna, Calculus
cursor.execute("INSERT INTO Results (student_id, exam_id, marks_obtained, grade) VALUES (?, ?, ?, ?);", (1, 2, 78, 'B')) # Anna, Programming

# Insert results for Ben Carter (student_id 2)
cursor.execute("INSERT INTO Results (student_id, exam_id, marks_obtained, grade) VALUES (?, ?, ?, ?);", (2, 1, 70, 'B')) # Ben, Calculus
cursor.execute("INSERT INTO Results (student_id, exam_id, marks_obtained, grade) VALUES (?, ?, ?, ?);", (2, 3, 65, 'C')) # Ben, Physics

# Insert results for Chloe Davis (student_id 3)
cursor.execute("INSERT INTO Results (student_id, exam_id, marks_obtained, grade) VALUES (?, ?, ?, ?);", (3, 2, 92, 'A')) # Chloe, Programming

conn.commit()
conn.close()

print("Examination results inserted.")
```

Examination results inserted.

This query joins the `Students`, `Results`, `Exams`, and `Subjects` tables to display all marks obtained by a specific student for each subject and exam type. The `student_id` can be changed to query for other students.

```
[53]
✓ Os
import sqlite3

conn = sqlite3.connect('university_examination.db')
cursor = conn.cursor()

cursor.execute('''
SELECT
    S.first_name || ' ' || S.last_name AS StudentName,
    SUB.subject_name,
    E.exam_type,
    R.marks_obtained,
    R.grade
FROM Students AS S
JOIN Results AS R ON S.student_id = R.student_id
JOIN Exams AS E ON R.exam_id = E.exam_id
JOIN Subjects AS SUB ON E.subject_id = SUB.subject_id
WHERE S.student_id = 1
ORDER BY SUB.subject_name, E.exam_date;
''')

results = cursor.fetchall()
conn.close()

print("Subject-wise marks for Student Anna Lee (ID=1):")
if results:
    for row in results:
        print(row)
else:
    print("No results found for this student.")
```

```
... Subject-wise marks for Student Anna Lee (ID=1):
('Anna Lee', 'Calculus I', 'Midterm', 85, 'A')
```

3. Calculate average marks for each subject

This query calculates the average `marks_obtained` for each `subject_name`, regardless of exam type. It joins `Results`, `Exams`, and `Subjects` tables, then uses `GROUP BY` and `AVG()`.

```
[54]
✓ Os
import sqlite3

conn = sqlite3.connect('university_examination.db')
cursor = conn.cursor()

cursor.execute('''
SELECT
    SUB.subject_name,
    AVG(R.marks_obtained) AS average_marks
FROM Results AS R
JOIN Exams AS E ON R.exam_id = E.exam_id
JOIN Subjects AS SUB ON E.subject_id = SUB.subject_id
GROUP BY SUB.subject_name
ORDER BY average_marks DESC;
''')

results = cursor.fetchall()
conn.close()

print("Average marks for each subject:")
if results:
    for row in results:
        print(row)
else:
    print("No results found to calculate averages.")
```

```
... Average marks for each subject:
('Introduction to Programming', 85.0)
('Calculus I', 73.33333333333333)
```